The empirical recurrence for OEIS A317562, recovered from its tail

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Abstract

OEIS A317562 records "Empirical recurrence of order 62" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 121$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 64$. Its last coefficient is $c_{62} = -1428352 \neq 0$, so no further tail satisfies anything shorter; any order-62 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A317562 is "Number of nX5 0..1 arrays with every element unequal to 0, 1, 2, 3, 4, 5, 7 or 8 king-move adjacent elements, with upper left element zero.". It has offset 1 and begins

$$16, 512, 11510, 255788, 5721314, 127754533, 2853437287, 63734296219, \ \dots$$

The entry states:

> Empirical recurrence of order 62 (see link above)

As of the "Last modified" line on the live entry (Jul 31 2018 12:36:49, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^\top M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 121$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 121$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 62: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 62.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 242$ exact terms from $n = 2$ on, it returns order 62 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{62} c_i\, a(n-i),$$

c_1	18	c_{17}	6306491139	c_{33}	−5355895990212	c_{49}	76210294810
c_2	183	c_{18}	42968313476	c_{34}	238438456610	c_{50}	−1572466842
c_3	−1002	c_{19}	97152554288	c_{35}	2749337225614	c_{51}	−615903049
c_4	−20365	c_{20}	97936180274	c_{36}	−1082854606034	c_{52}	15127095170
c_5	−37567	c_{21}	−169179457831	c_{37}	−1991505787877	c_{53}	6482885624
c_6	565063	c_{22}	−630082167490	c_{38}	491765019622	c_{54}	3256871952
c_7	3547716	c_{23}	−511758659821	c_{39}	776686281399	c_{55}	2198237101
c_8	352841	c_{24}	754527309454	c_{40}	123260909209	c_{56}	−569855468
c_9	−55616022	c_{25}	2163968673397	c_{41}	547316759121	c_{57}	−1199360516
c_{10}	−152363659	c_{26}	1210718263805	c_{42}	336419915839	c_{58}	−799450496
c_{11}	135149867	c_{27}	−2534023579471	c_{43}	−282294214357	c_{59}	−312687960
c_{12}	1355680510	c_{28}	−4703275417959	c_{44}	−193378205048	c_{60}	−76868976
c_{13}	1893618402	c_{29}	−922733717048	c_{45}	100619912777	c_{61}	−16313120
c_{14}	−2655275881	c_{30}	5492395160379	c_{46}	55013987429	c_{62}	−1428352
c_{15}	−11570618058	c_{31}	5431776140830	c_{47}	−30266130271		
c_{16}	−12188152023	c_{32}	−2127091803677	c_{48}	69251452078		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 64$, and it is the recurrence OEIS A317562 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{62} - c_1 x^{62-1} - \cdots - c_{62}$. Its constant term is $-c_{62} = 1428352$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 62 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 62, so $\deg q = 62$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 62, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 62, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1428352.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-62 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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